

Sales Forecast -- Monthly Retail Revenue

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Statistical Inference

Universe

- Stl Decomposition: $\{\text{trend_start, trend_end, trend_growth, seasonal_max, seasonal_min, seasonal_growth}\}$
- Sarima: $\{\text{order, seasonal_order, aic, bic, forecast, ci_lower, ci_upper, rmse, mape}\}$
- Holt Winters: $\{\text{forecast, sse, rmse, mape}\}$
- Residual Diagnostics: $\{\text{ljung_box_statistic, ljung_box_p_value, residuals_ok}\}$

Variables

Symbol	Meaning	Type / Range
μ_1, μ_2	population means for groups 1 and 2	$\mathbb{R}$
$\bar{x}_1, \bar{x}_2$	sample means	$\mathbb{R}$
s_1, s_2	sample standard deviations	$\mathbb{R}_{>0}$
n_1, n_2	sample sizes	$\mathbb{Z}^+$
t	test statistic (hypothesis test)	$\mathbb{R}$
p	p-value	$[0, 1]$
α	significance level	$[0, 1]$

Structure

A retail store has 4 years (48 months) of monthly revenue data with a clear upward trend and strong seasonal pattern (peak in December, trough in February). The owner needs to forecast revenue for the next 12 months to plan inventory, staffing, and cash flow.

The data exhibits three classic time series components:

- **Trend**: ~5% annual revenue growth driven by expanding customer base
- **Seasonality**: 12-month cycle with holiday spike (Dec) and post-holiday dip (Jan-Feb)
- **Noise**: Random month-to-month variation from weather, promotions, local events

Two complementary forecasting methods are applied:

1. **SARIMA(1,1,1)(1,1,1,12)** -- a parametric model capturing autoregressive and

moving-average structure in both the non-seasonal and seasonal components

2. **Holt-Winters (additive seasonal)** -- an exponential smoothing method that decomposes the series into level, trend, and seasonal components with exponentially decaying weights

Real-World Mapping

Real-World Concept

group average
group variability
sample count
significance threshold

Mathematical Object

sample mean $\bar{x}$
sample std s
 n
 $\alpha = 0.05$

Constraints

(1)

$$H_0 : \mu_1 = \mu_2$$

null hypothesis

(2)

$$H_1 : \mu_1 \neq \mu_2$$

alternative hypothesis

(3)

$$\alpha = 0.05$$

significance level

(4)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's t-test statistic

Objective

Reject H_0 if $p < \alpha$

Approach

Suggested approach SARIMA(1,1,1)(1,1,1,12) + Holt-Winters (additive seasonal)

Complexity class:

Available tools: SARIMA

2. Solution

Answer:	Statistical test complete
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	SARIMA(1,1,1)(1,1,1,12) + Holt-Winters (additive seasonal)
Time:	0.2749s
Certificate:	Assumptions checked; test statistic independently verified.

Solution Details

Statistical analysis computed.

Verification

✓ All Passed Passed

3. Interpretation

The Question

A retail store has 4 years (48 months) of monthly revenue data with a clear upward trend and strong seasonal pattern (peak in December, trough in February).

The Answer

Test complete.

What This Means

Stationarity Tests (raw data):

- ADF test: $p > 0.05$ (non-stationary, as expected with trend + seasonality)
- KPSS test: $p < 0.05$ (confirms non-stationarity)

After differencing ($d=1$, $D=1$, $s=12$):

- ADF test: $p < 0.05$ (stationary)

STL Decomposition:

- Trend: steadily increasing from ~\$100K to ~\$115K over 48 months
- Seasonal: December peak ~+\$15-20K, February trough ~-\$8-10K
- Residual: mean near zero, no obvious pattern

SARIMA(1,1,1)(1,1,1,12):

- AIC and BIC reported for model selection
- 12-month forecast with 95% confidence intervals
- Residuals pass Ljung-Box test (no significant autocorrelation)

Holt-Winters (additive):

- 12-month forecast with comparable accuracy
- Generally smoother forecast profile

Model Comparison:

- RMSE and MAPE on holdout (last 12 months)
- Both models achieve $\text{MAPE} < 15\%$
- AIC/BIC favor SARIMA for this data

Verification: All 7 checks pass (stationarity, positive forecasts, CI containment, residual diagnostics, MAPE threshold, seasonal period).

1. Generate synthetic monthly revenue data with trend, seasonality, and noise (48 months)
2. Test stationarity using Augmented Dickey-Fuller (ADF) and KPSS tests on the raw series
3. Decompose the series via STL (Seasonal-Trend decomposition using LOESS) to isolate trend, seasonal, and residual components
4. Difference the series (first difference + seasonal difference) to achieve stationarity
5. Fit SARIMA(1,1,1)(1,1,1,12) via maximum likelihood estimation and extract AIC/BIC
6. Fit Holt-Winters exponential smoothing with additive seasonal component
7. Forecast 12 months ahead with both models; compute 95% confidence intervals for SARIMA
8. Compare models on holdout data using RMSE and MAPE
9. Diagnose residuals via Ljung-Box test for remaining autocorrelation
10. Verify independently that all quality criteria are met

Recommendations

1. Report effect sizes alongside p-values for practical significance assessment.

Limitations

- Results are conditional on model assumptions (normality, independence, etc.).

• Statist